
CSS Basics

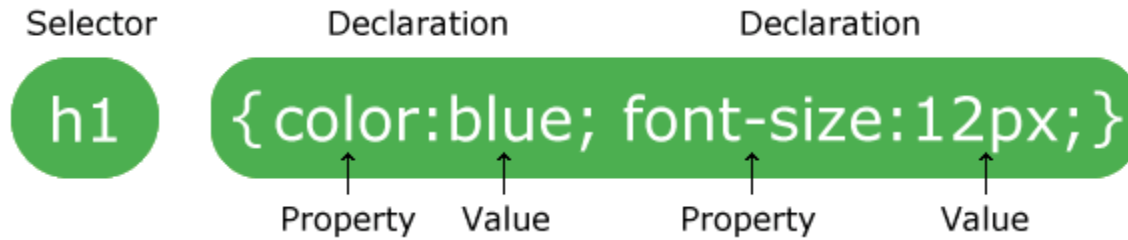
Ref: <https://www.w3schools.com/> and
other internet based tutorials

What is CSS?

- Using CSS (Cascading Style Sheets), we can apply styles to our web pages to make them look exactly how we want.
 - CSS describes how HTML elements are to be displayed on screen.
 - CSS saves a lot of work. It can control the layout of multiple web pages all at once.
 - External stylesheets are stored in CSS files.
 - CSS is used to define styles for your web pages, including the design, layout and variations.
 - The style definitions are normally saved in external .css files.
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CSS Syntax

- A CSS rule-set consists of a selector and a declaration block:



- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

```
p {  
  color: red;  
  text-align: center;  
}
```

CSS Selectors

- CSS selectors are used to select the HTML elements we want to style.
- We can divide CSS selectors into different categories:
 - Simple selectors (select elements based on name, id, class)
 - Combinator selectors (select elements based on a specific relationship between them)

```
div > p {  
    background-color: yellow;  
}
```

- Pseudo-class selectors (select elements based on a certain state)

```
a:hover {  
    color: #FF00FF;  
}
```

- Attribute selectors (select elements based on an attribute or attribute value)

```
input[type="text"] {  
    width: 150px;  
    display: block;  
}
```

- The CSS Universal Selector

- The universal selector (*) selects all HTML elements on the page.
- Example
- The CSS rule below will affect every HTML element on the page:

```
* {  
    text-align: center;  
    color: blue;  
}
```

- The CSS element Selector

- The element selector selects HTML elements based on the element name.
- Example
- Here, all <p> elements on the page will be center-aligned, with a red text color:

```
p {  
  text-align: center;  
  color: red;  
}
```

The CSS id Selector

- The id selector uses the id attribute of an HTML element to select a specific element.
- Example
- The CSS rule below will be applied to the HTML element with `id="para1"`:

```
#para1 {  
    text-align: center;  
    color: red;  
}
```

The CSS class Selector

- The class selector selects HTML elements with a specific class attribute.
- To select elements with a specific class, write a period (.) character, followed by the class name.
- In this example all HTML elements with class="center" will be red and center-aligned:

```
.center {  
  text-align: center;  
  color: red;  
}
```


The CSS class Selector

- You can also specify that only specific HTML elements should be affected by a class.

In this example only `<p>` elements with `class="center"` will be center-aligned:

```
p.center {  
  text-align: center;  
  color: red;  
}
```

- HTML elements can also refer to more than one class.
- In this example the `<p>` element will be styled according to `class="center"` and to `class="large"`:
- `<p class="center large">This paragraph refers to two classes.</p>`

The CSS Grouping Selector

- The grouping selector selects all the HTML elements with the same style definitions.
- Example
- In this example we have grouped the selectors:

```
h1, h2, p {  
  text-align: center;  
  color: red;  
}
```

Summary

Selector	Example	Example description
<u>.class</u>	.intro	Selects all elements with class="intro"
<u>#id</u>	#firstname	Selects the element with id="firstname"
<u>*</u> _	*	Selects all elements
<u>element</u>	p	Selects all <p> elements
<u>element,element,..</u>	div, p	Selects all <div> elements and all <p> elements

The Descendant Selector

- Descendant selectors let you apply styles to elements that are contained within other elements.
- For example, the following rule sets all text within `<b>...</b>` tags to red, but only if they occur within `<p>...</p>` tags (like this: `<p><b>Hello</b> there</p>`):
 - ❖ *`p b { color:red; }`*
 - ❖ *`ul li b { color:blue; }`*

The Child Selector

- The child selector is similar to the descendant selector but is more restrictive about when the style will be applied, by selecting only those elements that are direct children of another element.
- Consider `<p><i><b>Hello</b> there</i></p>`
- `p b { color:red; }` will change Hello to red, even if the bold text is itself within italics. While,
- `p > b { color:red; }` will not change the word Hello to red because it is not a direct child of the paragraph